

Extended Abstract: Interaction-Triggered Estimation of AR Object Placement on Indeterminate Meshes

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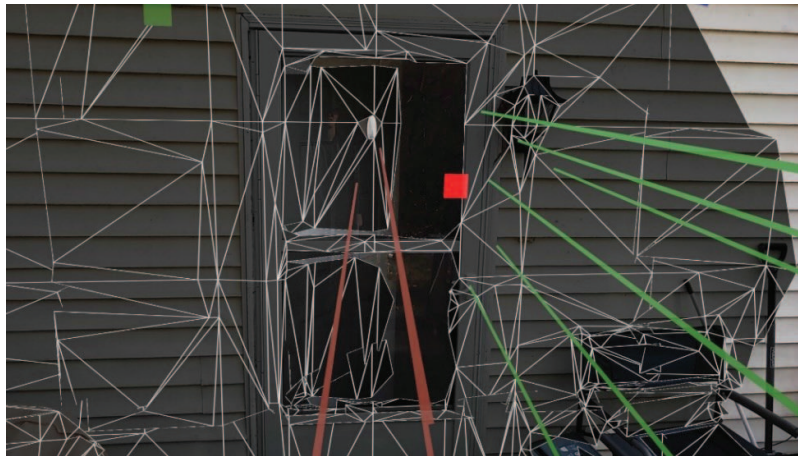


Figure 1: Hole in the environment mesh scan provided by the HoloLens 2 and the resulting object placement calculated by our proposed estimation solution

ABSTRACT

Current Augmented Reality devices rely heavily on live environment mapping to provide convincing world-relative experiences through user interaction with the real world. This mapping is obtained and updated through many different algorithms but often contains holes and other mesh artifacts when generated in less ideal scenarios, like outdoors and with fast movement. In this paper, we present the Interaction-Triggered Estimation of AR Object Placement on Indeterminate Meshes, a work-in-progress application providing a quick, interaction-triggered method to estimate the normal and position of missing mesh in real-time with low computational overhead. We achieve this by extending the user's hand using a group of additional raycast sample points, aggregating results according to different algorithms, and then using the resulting values to place an object.

Keywords: Augmented reality, user experience, head-worn display, mesh repair, mesh estimation.

Index Terms: Human-centered computing→Human computer interaction (HCI)→Interaction paradigms→Mixed / augmented reality

1 INTRODUCTION

Today's Augmented Reality (AR) has slowly migrated from small, controlled, room-scale experiences to outdoor, world-scale

experiences. This expansion has seen impressive, extensive AR experiences used increasingly in commercial and consumer sectors, such as Hansen et al.'s [1] AR subsurface utility engineering visualization for civil engineers and Blanco-Novoa et al.'s [2] augmented industrial engineering visualization for shipyards. These large AR experiences have strained many modern AR devices' standalone tracking and mapping systems, like the HoloLens 2, causing them not to perform optimally. Researchers found that the HoloLens 2 SLAM algorithm produced tracking and mapping results within a few centimeters of the ground truth building model in single room-scale tests, but quickly saw large deviations from ground truth with multiple connected rooms [3].

While much research focuses on the localization problem presented with large-scale AR tracking, we believe equally crucial to the end-user experience is the quality of the environment mapping done by the device. This environment mapping is primarily what the user interacts with to place and manipulate AR objects in their real-world environment. Environment mapping is usually represented and stored as 3D triangular mesh, with the most common problem encountered being holes in the mesh. These mesh holes are caused by many issues like sunlight, awkward environment features (plain white wall with no textures), or improper scanning by the user, and cause unwanted user problems when they are in areas with which the user wishes to interact. Most AR environment mesh improvement plans aim to modify the entire mesh generation algorithm in order to generate a near "perfect" mesh. This "perfect" mesh would ideally be a one-to-one digital twin of the real-world environment. While good in practice, we believe this method is far too complicated and prone to diminishing returns in the long run, in addition to not working for older mapping systems or systems with limited computing onboard.

Traditional mesh repair could also be considered a more logical solution to this problem. Many mesh repair algorithms exist, such as surface inpainting [4], but these algorithms are geared toward repairing 3D models and, in most cases, run on desktop computers

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with much higher computing power. Additionally, environment mesh is often not organized well enough in the AR device to fix holes locally without considering the entire mesh at all times, leading to a high computational load if needed to update constantly.

2 PROPOSED SOLUTION

We propose a simple work-in-progress approach to dealing with the holes and inconsistencies present in many AR environment meshes by utilizing an interaction-triggered repair technique based on multiple raycast sampling. We attach a virtual plane, seen in Figure 2, normal to the palm of the user's hand, which extends the user's AR interaction pointer using a grid of sampling points. When the user encounters a hole in the environment mesh, the system:

1. Samples the mesh around the hole using the grid of sampling points to collect distances and mesh normals
2. Applies outlier elimination to all of the returned sample points using modified z-scoring [5]
3. Aggregates the remaining distances and normals according to one of three algorithms found in Table 1
4. Places item in the estimated position utilizing the initial interaction vector and resulting algorithm return values

Table 1: Different Implemented Depth and Normal Aggregate Algorithms

Technique	Calculation Method
Pure Average	All sample points are aggregated equally
Weighted Average	Sample points originating closer to the original hand interaction ray are weighted higher
Eye-weighted Average	Sample points hitting the mesh closest to the user's current gaze are weighted higher

Our solution does not require constant mesh repair code to be run, rather it only uses computational resources when the user intends to interact with the environment. This allows for more compute overhead for the actual AR application and ensures no time is wasted repairing areas that may be fixed by the mapping system itself in the future. Additionally, this concept and codebase are very architecture agnostic as it uses basic building blocks from any game engine and utilized standard math formulas.

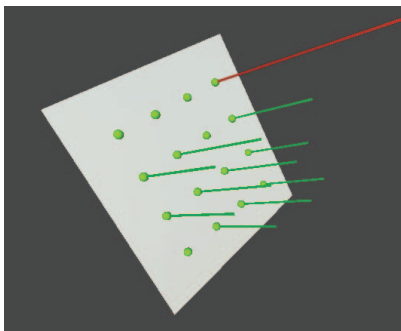
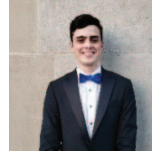


Figure 2: Hand plane attached to user's palm containing sampling points

3 NEXT STEPS

We plan to continue internally testing the different combinations of depth and normal calculation methods presented, finding the best solution for different scenarios. We then plan to run an outdoor user study later this year to determine the effectiveness and usability of our solution.

4 ABOUT THE AUTHORS



John Luksas is a current master's student at the Virginia Tech Grado Department of Industrial & Systems Engineering. He is part of the Cognitive Engineering for Novel Technologies (COGENT) Lab, where his research interest lies in the expansion of reliable outdoor AR and interaction methods.



Joseph L. Gabbard received his MS and Ph.D. degrees in computer science from Virginia Tech. He is the director of the Cognitive Engineering for Novel Technologies (COGENT) Lab and an associate professor of Human Factors at Virginia Tech's Grado Department of Industrial & Systems Engineering. He is also an executive committee member of Virginia Tech's Center for Human-Computer Interaction. His research focuses on the connections between user interface design and human performance, and specifically the development of techniques to design and evaluate AR and VR user interfaces.

ACKNOWLEDGMENTS

This material is based upon work supported by the National Institute of Standards and Technology, as well as the Virginia Transportation Research Council.

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